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Modeling of gravity-wave tail spectra in the middle atmosphere via numerical and Doppler-spread methods

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Abstract

Wind fluctuations in the middle atmosphere exhibit an Eulerian spectral tail approximating to the form m^{-3} at large vertical wavenumbers m. The tail is taken here to result from a characteristically Eulerian nonlinearity in the governing fluid dynamic equations, a nonlinearity that is absent from a Lagrangian description. It is modeled here via a previously described analytic transformation from assumed model Lagrangian input spectra. The present work gives numerical expression to the earlier analytic results for the intensity of the tail. The resultant tail spectra are found to represent well the relevant data at reasonable values of Richardson number. Under an assumption of constant Richardson number, they exhibit saturation as height increases, as do the observations. Previously obtained approximate results, including those of the Doppler-spread theory of saturation, are found to represent well the new numerical results. Implications for that theory and for the associated Doppler-spread parameterization are discussed in a companion paper.
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1. Introduction

Gravity waves in the middle atmosphere, as revealed by the irregular horizontal winds they are believed to produce, tend to exhibit a spectral tail at large vertical wavenumber m. This tail has the form ZN^2m^{-3}, where N is the background buoyancy frequency and Z is a coefficient of order 0.2 when N and m are defined in radian units. The coefficient tends to be invariant under wide changes of circumstance, and in particular appears to be independent of height despite the growth of amplitude with height that might be expected to accompany the decrease of gas density (e.g., Allen and Vincent, 1995; Dewan and Good, 1986; Smith et al., 1987; Tsuda et al., 1989). It is widely held that this behavior is indicative of some saturation process at work in the formation of the tail, but to date there exists no consensus as to the operative mechanism (e.g., Dunkerton, 1989; Fritts, 1989; Fritts and Alexander, 2003; Gardner, 1996; Hines, 1998).

The view adopted here is that, from an Eulerian perspective, the m^{-3} form is produced as a consequence of direct wave–wave interaction introduced mathematically via the nondissipative $\mathbf{v}\cdot\nabla$ nonlinearity of the Eulerian governing equations, while the magnitude of Z is determined by a requirement that the wave system as a whole should maintain itself marginally unstable as it propagates upward. This view was first set out, and was developed mathematically, on the somewhat heuristic basis that the nonlinear effects would be revealed statistically by a Doppler spreading of the intrinsic frequencies and vertical wavenumbers of the incident spectrum (Hines, 1991, 1993). The resultant Doppler-spread theory (DST) was subsequently adapted to provide a Doppler-spread parameterization (DSP) for the deposition of wave-borne momentum and energy in the middle atmosphere (Hines, 1997a, b), and so has entered numerical modeling of the large-scale circulation of the middle atmosphere (e.g., Akmaev, 2001; Lawrence, 1997; Manzini et al., 1997; Mayr et al., 1998). The validity or invalidity of the DST and the DSP is a matter of some importance for this reason.

Both the DST and the DSP remain suspect because of the heuristic origins of their mathematical development. From

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the outset, the far more rigorous nonlinear analysis presented by Allen and Joseph (1989, henceforth AJ89), to be outlined shortly, was acknowledged to be superior in principle even if far more difficult in development and application. The analysis by Chunchuzov (1996), which adapted AJ89 from its original oceanic context into an atmospheric context, is similarly far more rigorous and far more difficult. Neither the DST nor the DSP can claim validity, even as an approximation, unless the conclusions they produce are approximately consistent with those that might be developed from the more rigorous approach. Their potential in this regard is put to the test here.

The very difficult AJ89 approach has been rendered more accessible to the average reader—to any not versed in correlation theory and its application to spectra—, has been corrected in one respect, and has been extended to general input spectra for various observable parameters by Hines (2001, henceforth H01). The relations for Z found in H01 were so complex in form that only approximate numerical values could be determined there, but those values exhibited sufficient agreement with the observation (at reasonable Richardson numbers) that full numerical computations appeared to be warranted. We present here the results of some sample computations, and we compare them with the approximate versions that were previously available.

The formal basis of the AJ89 approach lies in a difference of description in Eulerian and Lagrangian coordinates. At sufficiently small amplitudes, a system of waves can be described just as simply in Eulerian coordinates as in Lagrangian coordinates; but, with growth of amplitude (with height, in the atmosphere), in the most relevant circumstances, the Eulerian description becomes thoroughly nonlinear and essentially intractable, whereas the Lagrangian description remains quasi linear (see AJ89 or the more explicit analysis by Hines, 2002, henceforth H02). As AJ89 showed, in such circumstances, the spectrum of the Eulerian description can still be inferred from the spectrum of the Lagrangian description, which itself remains unchanged in form unless there are selective dissipative losses that accompany the growth of amplitude. The transcription would require massive computations if conducted for the entire spectrum, but these reduce to a much more workable level if attention is confined to the large-m Eulerian tail. In the end (but see Appendix A), this tail is found to have the form m^{-3} as is observed, and its intensity is computable from the formulae of H01 for a wide range of arbitrarily chosen Lagrangian spectra.

Chunchuzov (2002, henceforth Ch02) has presented an analysis that is similar in principle to that of H01. It includes, in particular, the same basic correction to AJ89 (and likewise to Chunchuzov, 1996) as that reported in H01. It deals in part with somewhat different aspects of the problem, and it is constrained to a specific analytic form for the input spectrum. The two analyses are nevertheless fully concordant.

Here we employ the formulae of H01, but with the correction of a typographical error that should be noted: in its Eqs. (E.2) and (E.3), the k^2 that appears three times without subscripts should be κ^2. We adopt an input spectrum that has been designed to represent (crudely) the spectrum in the middle atmosphere, and we present the coefficient Z that results from numerical integrations and analytic extension. The input spectrum is set out in Section 2. In Section 3, we illustrate the output Eulerian tail spectra that are obtained for horizontal and vertical velocity and for horizontal and vertical displacement, at a selection of input intensities, and we compare the results with those obtained via the rough analytic evaluation given in H01 and with those found observationally.

We go on, in Section 4, to describe the anticipated dissipation and so to illustrate the expected variation with height of the m^{-3} Eulerian tail. The tail is found to be essentially unchanging in intensity with height. Thus, saturation is found to have been produced, just as it is observed in practice. This saturation is shown (analytically) to recur in a broad class of input spectral forms.

In Section 5, we compare the degrees of nonlinearity implicit in both the Eulerian and the Lagrangian spectra, the former being substantially the greater. We believe that this justifies our use of linear Lagrangian relations to infer the highly nonlinear Eulerian spectra.

In Section 6, we compare the computed tail spectrum with the spectral form produced by the DST, and find that the DST closely represents the spectrum it was intended to represent, over the relevant ranges of intensity and wavenumber. Discussion of this result, and of its implications, is to be found in a companion paper (Hines, 2004). Section 7 expands upon our conclusions in the more general context of spectral-tail theories.

2. The Lagrangian input spectrum

The relations of H01 are expressed in terms of an input Lagrangian power spectral density (PSD) of vertical displacement s_3, whose form must be assumed. This form is to be given as a function of the wavenumber vector (or wavevector) $\boldsymbol{\alpha}$, its variation with frequency $\beta(>0)$ being implied by the linear dispersion relation. It is taken to be azimuthally isotropic, and so becomes a function only of horizontal wavenumber $\alpha_h(>0)$ and vertical wavenumber α_3. Background winds and Coriolis effects are ignored. Dissipation is likewise ignored, except as it is introduced "by hand" in Section 4.

We have arbitrarily adopted the input spectral form

$$\mathrm{PSD}_{s_3} = [A/2\pi]\,\alpha_h^{-3}\,\alpha_3^2 \tag{2.1}$$

over a finite range of α_h and α_3 soon to be described, and we have taken it to vanish outside that range. Here A is an adjustable constant (having dimensions [length]^4) that can be used to vary the intensity of the wave system. This PSD is defined in the sense that, for analytic purposes, it will

produce the variance of s_3 if multiplied by $2\pi\alpha_h d\alpha_h d\alpha_3$ and integrated.

The range of input wavenumbers is limited by

$$|\alpha_3|\beta_m/N \leqslant \alpha_h \leqslant |\alpha_3|\beta_M/N \tag{2.2}$$

and

$$\alpha_m \leqslant |\alpha_3| \leqslant \alpha_M. \tag{2.3}$$

Here, α_m is a minimum vertical wavenumber taken to be (2π radians)/(60,000 m) throughout our illustrations; α_M is a selectable maximum vertical wavenumber, taken to be (2π radians)/(1500 m), (2π radians)/(3000 m) and (2π radians)/(6000 m) in various results to be presented; N is the buoyancy frequency, taken to be (2π radians)/(360 s) throughout the computations; β_m is a minimum wave frequency, taken to be $N/24$ throughout, and β_M is a maximum wave frequency, taken to be $N/8$ throughout, with corresponding periods 2.4 h and 48 min, respectively. The limits in Eq. (2.2) are functions of $|\alpha_3|$ designed for compatibility with the approximate gravity-wave dispersion relation

$$|\alpha_3| = \alpha_h N/\beta, \tag{2.4}$$

which is assumed in H01 and so also here. The extreme limits are such that horizontal wavelengths lie within the range 12–1440 km. In most results to be presented, we adopt solely upgoing waves, for which α_3 is negative. In one case—that is illustrated first, in Fig. 1a —we adopt vertically symmetric spectra (as appears to be appropriate for most oceanic spectra), with an intensity factor A' given by half the A of the corresponding one-sided input spectrum in order to produce equal total energy densities and variances. In this case, the positive-α_3 and negative-α_3 portions of the spectrum are combined, when illustrated, to produce a composite $|\alpha_3|$ ($=m$) spectrum such as observations typically provide.

No claim is made that the adopted spectrum constitutes a particularly good representation of the actual middle-atmosphere s_3 Lagrangian spectrum, but it does not have to in order to produce representative results. This is because, in the end, the Eulerian results depend almost exclusively on a Lagrangian Richardson number Ri_L, soon to be defined, and our results will employ Ri_L values that both illustrate the processes at work and lead to inferences appropriate to the middle atmosphere. The form adopted in Eq. (2.1) is just a convenient form for purposes of determining certain spectral integrals that are required in order to get on with the computations. It was adopted in advance of any numerical work, and has not been redesigned in any way to make any particular point other than what the numerical results themselves might produce. Readers may freely choose alternative spectra for their own computations, but they should expect to find results for the coefficient Z that closely match those obtained here for the same Ri_L. This fact has physical implications, discussed in Section 4, in relation to saturation.

The analysis in H01 was conducted for vertical wavelengths not greatly in excess of an atmospheric scale height. We breach this condition somewhat, at the larger wavelengths illustrated here, but the relevant portion of the spectrum is only a weak contributor to the end results and would remain so even if a more complete analysis were employed.

We shall be presenting vertical-wavenumber (m) spectra for the total horizontal velocity v_h (with "total" implying that the spectra for velocity components in two orthogonal horizontal directions have been combined), for the vertical velocity v_3, for the total horizontal displacement s_h and for the vertical displacement s_3. Both the input Lagrangian spectra and the corresponding output Eulerian spectral tails will be shown.

The respective input Lagrangian m spectra for the various wave fluctuations are derived from Eq. (2.1) with the aid of approximate gravity-wave polarization relations, taken from H01 and reasonably valid within our chosen ranges of β and m. These imply, for example, that the amplitude factor V_h for the horizontal velocity induced by an individual wave is given by N times the amplitude factor S_3 for vertical displacement induced by the same wave, hence the PSD of v_h is given by N^2 times the PSD of s_3. Similarly, to obtain the input PSD of s_h, the spectrum (2.1) must be multiplied by $(\alpha_3/\alpha_h)^2$ before integration over α_h, and to obtain the input PSD of v_3 it must be multiplied by $N^2(\alpha_h/\alpha_3)^2$ before integration over α_h. In all cases, the result of the integration is an input m spectrum proportional to m over the range $\alpha_m < m < \alpha_M$, and 0 outside this range. For example, within that range the s_3 spectrum found upon integration over α_h is $16Am$, and the v_h spectrum is $16AN^2m$. A form proportional to m is commonly adopted for the Eulerian spectrum of v_h in the middle atmosphere, at wavenumbers less than some value (usually denoted m_*) that marks the onset of the tail, and so is a suitable form for the input Lagrangian m spectrum in the present model.

H01 employs several "spectral sums", of which the most important is

$$I_1 = (1/2)\sum S_3^2\alpha_3^2 = (1/2N^2)\sum V_h^2\alpha_3^2, \tag{2.5}$$

where the summation is over all contributing individual waves. The final expression, which is obtained by application of a standard gravity-wave approximation once again, reveals that I_1 is not only the variance of vertical gradient of vertical displacement but also the variance of vertical shear of horizontal velocity, normalized by the unperturbed N^2. It is therefore the inverse of a form of Lagrangian Richardson number Ri_L. For the present application, we rewrite half the sum as a variance obtained by integration over our spectrum, employing the middle expression in Eq. (2.5) to produce

$$\begin{aligned} Ri_L^{-1} = I_1 &= \iint \mathrm{A}\alpha_h^{-2}\alpha_3^4\, d\alpha_h\, d\alpha_3 \\ &= (\mathrm{AN}/4)(\alpha_M^4 - \alpha_m^4)(\beta_m^{-1} - \beta_M^{-1}) \\ &= 4\mathrm{A}(\alpha_M^4 - \alpha_m^4). \end{aligned} \tag{2.6}$$

Here, the limits of integration are as already described in Eqs. (2.2) and (2.3); the α_h integration is performed first, since its limits depend on $|\alpha_3|$, and the final expression is to be evaluated for the currently specified value of α_M. Since all $\alpha_M^4 \gg \alpha_m^4$ under our current specifications, we can obtain

$$Ri_L = [4A\alpha_M^4]^{-1} \tag{2.7}$$

approximately. All other spectral sums are similarly insensitive to the value of α_m so long as it is $\ll \alpha_M$. This fact lends credence to our incorporation of the larger vertical wavelengths employed here while neglecting corrections associated with the atmospheric scale height, since these wavelengths could be omitted without significant change. The results to be presented here were derived for five values of A taken in equal logarithmic steps in the range $(180\alpha_M^4)^{-1} \leqslant A \leqslant (18\alpha_M^4)^{-1}$, and so for Ri_L ranging from 45 down to 4.5. The latter limit lies close to the edge of applicability of the H01 analysis (see Section 5). Indeed, the accuracy of our numerical results, as being truly representative of the Eulerian spectrum, becomes increasingly suspect as this final value of Ri_L is approached, but the validity of the results as an indicator of the increasing consequences of Eulerian nonlinearity is not to be denied.

Other spectral sums required by the H01 analysis were evaluated as spectral integrals in a similar fashion. As examples, for $\alpha_M = (2\pi \text{ radians})/(1500 \text{ m})$ and $A = (18\alpha_M^4)^{-1}$ the values found were: $I_1 = 0.22222$, $I_2 = 1.15741 \times 10^{-3}$ (=variance of horizontal shear of vertical displacement), $I_3 = 61.6296$ (=variance of vertical shear of horizontal displacement), $I_4 = 7.7112 \text{ m}^2 \text{ s}^{-2}$ (=variance of horizontal velocity), $I_5 = 4.0163 \times 10^{-2} \text{ m}^2 \text{ s}^{-2}$ (=variance of vertical velocity), $I_6 = -19.7528 \text{ m s}^{-1}$, $I_7 = -8.47681 \times 10^{-2} \text{ m s}^{-1}$, $I_8 = 1.01853 \times 10^{-10} \text{ m}^{-2}$, $I_9 = 1.35386 \times 10^{-8} \text{ m}^{-2}$, $I_{10} = 2.59940 \times 10^{-6} \text{ m}^{-2}$, $I_{11} = 7.20901 \times 10^{-4} \text{ m}^{-2}$, $I_{12} = -5.64106 \times 10^{-9} \text{ m}^{-1} \text{ s}^{-1}$, $I_{13} = -8.92417 \times 10^{-7} \text{ m}^{-1} \text{ s}^{-1}$, $I_{14} = -2.07952 \times 10^{-4} \text{ m}^{-1} \text{ s}^{-1}$, $I_{15} = 2.65241 \times 10^{-9} \text{ s}^{-2}$, $I_{16} = 2.5314 \times 10^4 \text{ m}^2$ (= variance of vertical displacement) and $I_{17} = 7.0205 \times 10^6 \text{ m}^2$ (= variance of horizontal displacement). The negative signs arise from the case of solely upgoing waves. For solely downgoing waves, the negative signs would be replaced by positive signs; but they always enter the formulae as products of pairs, so the end result would be the same. For spectra containing the same total energy density but symmetric upgoing and downgoing components, the negative values would be replaced by 0 while the others would remain unchanged. For the same α_M but varying A, all are proportional to A and so remain in the same proportions relative to one another.

The spectral sums have been found to present a remarkable pattern of behavior. If an initially adopted 3D input PSD for s_3 were to be replaced by another found by mapping, such that the spectral density found initially at α_h, α_3 is now to be found at $f\alpha_h, f\alpha_3$, save for an increase or decrease of overall intensity to maintain Ri_L constant, then the ratios $I_1 : I_2 : I_3 : \ldots$ will vary as $1 : 1 : 1 : f^{-2} : f^{-2}; f^{-1} : f^{-1} : f^2 : f^2 : f^2 : f^2 : f : f : f : 1 : f^{-2} : f^{-2}$. The mathematical significance of this behavior lies in the fact that these spectral sums enter the H01 formulae for Z only in distinctive clusters (e.g., $I_1, I_2, I_6 I_{12}, I_7^2 I_{10}$), and all these remain quite invariant under the change. In our case, the scaling could be achieved by simultaneous scaling of α_m and α_M, and then of A in keeping with Eq. (2.7). Thus, while we present our spectra for the values already stated, they can be found for other values (with α_m/α_M and Ri_L held constant) simply by shifting the abscissa scale by any chosen factor and the ordinate scale by that factor's inverse cube. (The actual wavenumbers employed in our computations were larger by a factor of 10 than those illustrated here, in order to fully eliminate any complications that might be attributed to finite atmospheric scale height. The results displayed here are scaled from those computed.) The physical significance of this remarkable behavior will be made apparent in Section 4, where the saturation of wave spectra is discussed.

3. Sample *m* Eulerian spectra of v_h, v_3, s_h and s_3

Relations (D.l)–(D.5) and (E.1)–(E.5) (corrected as stated in Section 1 here) and Table 2 of H01 were employed in the manner described there to provide the lowest-order asymptotic term of the Eulerian tail spectra for the horizontal and vertical velocity v_h and v_3 and for the horizontal and vertical displacement s_h and s_3. This asymptotic term was found, for each variable in turn, first for the azimuthally isotropic 3D spectra that the analysis produces, and then, via numerical integration over horizontal wavenumber, for the 1D single-sided m spectrum. The resultant asymptotic term, which is proportional to m^{-3}, is the only relevant term at large m and the only one available from H01 (or from AJ89 or from Ch02, for that matter). We present the results graphically, first in Fig. 1 for the prototypical case of v_h when $\alpha_M = (2\pi \text{ radians})/(1500 \text{ m})$.

Fig. 1a displays, in its upper-left quadrant, five input m spectra for v_h, obtained for $Ri_L = 45.0$, 25.3, 14.2, 8.0 and 4.5 (from the bottom to top). These values represent successively increasing levels of wave intensity in equal logarithmic steps. All input spectra drop discontinuously to 0 at their lower and upper limits, as is represented by the vertical lines. To their right are displayed the corresponding m^{-3} asymptotic terms of the one-dimensional Eulerian output spectra.

As will be seen, the nonlinear Eulerian asymptotic term is essentially of negligible magnitude when Ri_L is as large as 45. But, with an increase of input wave intensity by a single factor of 10, leading to a decrease of Ri_L to 4.5, the tail increases in intensity by more than six orders of magnitude and at its small-m end reaches intensities comparable to those of the input spectrum. (The asymptotic term proper loses mathematical validity as m decreases toward this lower end, which we have taken arbitrarily as the central vertical

line. Extrapolation into this region from larger m becomes a matter of judgement, to be given below when the marriage of the Eulerian tail spectrum to the Lagrangian input spectrum is discussed. For the moment, we simply depict the form and intensity of the term itself.)

The analysis of H01, with its rough estimate of the integrations that have here been done numerically, yielded for v_h the 1D asymptotic term

$$\mathrm{PSD}_{v_h\ \mathrm{tail}} = Z_a N^2 m^{-3} \tag{3.1}$$

with Z_a an approximation to the true Z given (for radian units) by

$$Z_a = (32\pi)^{-1/2} Ri_L^{3/2} \exp[-Ri_L/2]. \tag{3.2}$$

(Ch02 obtained the same approximation with his choice of spectra.) Had Z_a been a correct estimate, the various asymptotes would have passed through the respective dots that are superimposed on the graph. It will be seen that the rough estimate is reasonably reliable, certainly as to the general form of variation with Ri_L and even as to individual magnitudes.

Fig. 1a is drawn for the case of vertically symmetric spectra, such as would produce vertically standing waves in the atmosphere (or in the ocean). All further graphs presented in this paper are drawn for the case of waves propagating upward only, as might better represent much of the middle atmosphere. The two cases differ because phase relationships between v_h and the wave-induced displacement differ, and the difference is important to the correlations that are required. (See Eckermann (1999) for an illustration of the effect, or H01 for the mathematics.)

Fig. 1b gives results parallel to those of Fig. 1a, but now for the vertically asymmetric case having the same total energy density. The rough estimate (3.2) is unable to make any distinction between the two cases, so the dots that represent it lie in the same positions as before. They may be employed as a convenient means of comparing the tail spectra of the two cases. As will be seen, the general pattern of behavior is the same for the two cases, though at the smaller Ri_L values the tails in Fig. 1b are about four times as strong as those in Fig. 1a. Most of this difference can be traced to the $\widehat{F}$ term in Eq. (D.3) of H01, a term that has no counterpart in AJ89. If it were arbitrarily eliminated from the numerical computations, the figure corresponding to Fig. 1a would remain virtually unchanged and that corresponding to Fig. 1b would differ only slightly from Fig. 1a.

The term in $\widehat{F}$, which is of a complicated form, follows from the more basic correction to AJ89 mentioned above. It was not included by Ch02 as a matter of convenience, in the belief that it would not alter the end results by an order of magnitude (Chunchuzov, 2000, personal communication). The present computations reveal that this belief was well founded in the case of vertically symmetric spectra, but only just valid—the factor of change being about 4—for vertically asymmetric v_h spectra at the smaller Ri_L values. In the absence of this factor and of other terms that improve upon the preliminary approximation (3.2), that approximation yields a Z_a that never exceeds 0.12 (not 0.14 as stated on p. 1764 of Ch02). It seems to be this value on which Fritts and Alexander (2003) based their objection (para. 122) that Hines (2001) and Chunchuzov (2002) obtained a tail intensity about 2 to 3 times smaller than typically observed. As will be seen now, and as will be discussed below, that objection does not hold true when the full evaluation of tail intensity is employed.

Figs. 1a and b simply give a graphical representation of the asymptotic m^{-3} term of the Eulerian tail. What one would like to describe is the full Eulerian spectrum. This can in principle be done by numerical integrations of functional forms that are far more complicated than those required for the asymptotic term alone, forms given in H01 by insertion of its Eq. (8.1) or Eq. (8.2) into its Eq. (5.2). Any such effort would probably be wasted, if only because the input spectra are so poorly known in practice. We therefore proceed instead via inference and interpolation.

A crucial fact known about the Eulerian spectrum (AJ89, H01) is that, at vertical wavenumbers substantially less than the inverse of the rms vertical displacement $\langle s_3 \rangle$, the Eulerian spectrum should differ only marginally from the input Lagrangian spectrum. (This is so, at least, if the bulk of the wave energy lies there and at horizontal wavenumbers substantially less than the inverse of the rms horizontal displacement $\langle s_h \rangle$.) We therefore exhibit on each input term of Fig. 1b a marker, seen just to the left of the central vertical line, placed arbitrarily at $m = 0.25\langle s_3 \rangle^{-1}$. To the left of each marker, the Eulerian spectrum may be taken to be indistinguishable from the Lagrangian spectrum. The absence of such a marker in the case $Ri_L = 45$ implies that the Eulerian spectrum is virtually identical to the Lagrangian spectrum over the entire range of the latter. As Ri_L decreases from 45, a meaningful range of wavenumbers comes to be affected by the transition from the Lagrangian spectrum to the Eulerian. To depict the full Eulerian spectra, we adopt the Lagrangian spectra at small m, up to the point where a marker is reached, and then link them in some reasonable manner to their respective Eulerian asymptotic tail spectra.

A further fact constrains this linkage. The variance of velocity is a physical property whose evaluation should not, it might be thought, be dependent on the coordinate system being employed. On this basis, the area under the Eulerian spectral curve might be expected to match that under the Lagrangian spectral curve, when both curves are replotted (if only mentally) in linear rather than log–log coordinates. Such an inference would be entirely correct if the medium were quite incompressible, though it would require modification if compressibility were admitted (through use of the Jacobian of the coordinate transformation as a weighting factor), as has been shown rigorously by one of us (see Appendix B). Since the analysis in H01 was based on an assumption of incompressibility, and since (given the crudity of the model spectrum) we see no point in actually measuring areas for the present purpose, we may simply sketch transitions from the small-m Lagrangian spectra to

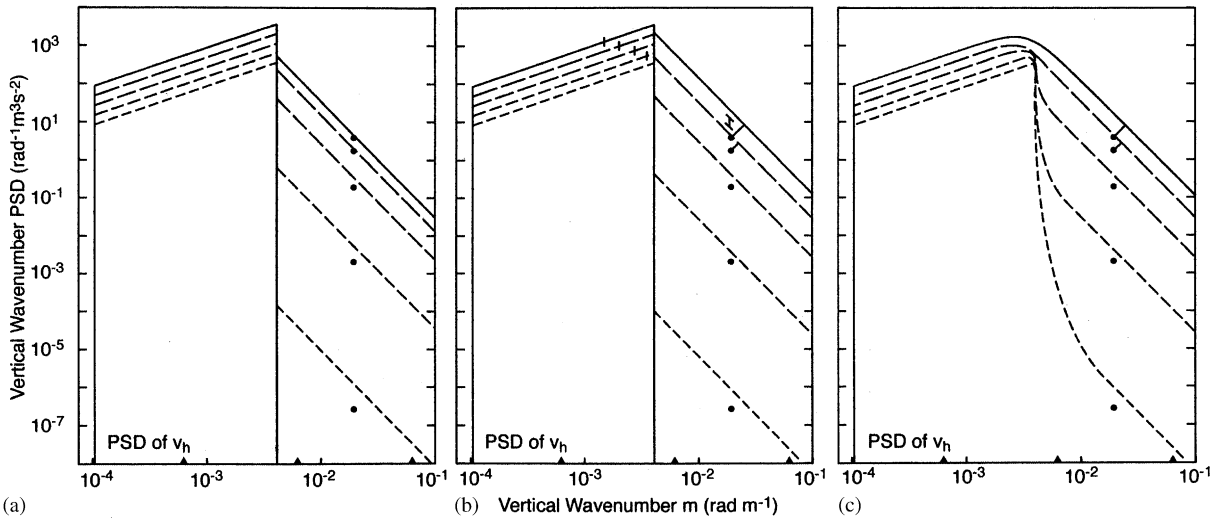

Fig. 1. Model vertical-wavenumber (m) spectra of horizontal velocity v_h, obtained for Lagrangian Richardson numbers $Ri_L = 45.0$, 25.3, 14.2, 8.0 and 4.5, respectively (from bottom to top). Panel (a) is drawn for vertically symmetric input spectra, shown on the left with sudden cutoffs represented by the vertical lines. Shown on the right are the corresponding output m^{-3} asymptotic terms of the nonlinear Eulerian tail, together with dots through which the lines would have passed if the analytic approximation (3.2) had been accurate. Panel (b) is drawn similarly for the same input spectral form but with only upward (or downward) propagating waves, with the same total energy density. The dots have the same interpretation as before and, being identical to those in panel (a), permit the two sets of output asymptotic terms to be compared readily. The short vertical bar is representative of the range of intensities reported for the observed m^{-3} tail in the middle atmosphere. The ticks on the input spectra mark m values equal to $0.25\langle s_3\rangle^{-1}$, to the left of which the full Eulerian spectra must closely match the input Lagrangian spectra. Panel (c) represents the full Eulerian spectra inferred for the various Ri_L under constraints described in the text. The spectrum drawn for $Ri_L = 4.5$ has been taken to be a standard model spectrum described in Section 7. The filled triangles along the abscissa represent vertical wavelengths of 10,000, 1000 and 100 m, respectively.

the large-m asymptotic tails in such a way as would seem to meet the constraint. Samples of such transitions are exhibited in Fig. 1c. The full curves seen there then represent the full Eulerian spectra to which the Lagrangian spectra of Fig. 1b would give rise. There would appear to be little room for modification of the forms shown, particularly at the smaller values of Ri_L. For them, one may reasonably infer that the asymptotic term proper remains approximately valid even as m decreases into the realm where the asymptotic analysis that produced it loses its strict validity.

It is only at the smaller values of Ri_L—at 4.5 and 8—that the resultant spectra tend to duplicate the forms of observed spectra (to the extent that they are indeed observed) or of standard model spectra. Accordingly, we anticipate that the naturally occurring spectral tails are to be associated with these smaller values of Ri_L. This will come as no surprise to those who believe that the tails are to be associated with similarly defined Eulerian Richardson numbers Ri_E of order 1 or less. The two Richardson numbers are by no means equivalent, and small Ri_E by no means implies small Ri_L automatically (see Section 4), but we will in fact confine our attention in all subsequent sections to the smaller values of Ri_L.

Before proceeding to those sections, we present in Figs. 2a–c graphs corresponding to those of Fig. 1b, drawn now for s_h, s_3 and v_3, respectively. The input m spectra of s_3 have already been specified. Those of s_h are the same except for multiplication by I_3/I_1, and those for v_3 are the same except for multiplication by $N^2 I_2/I_1$. These factors arise from polarization factors already described and from subsequent integrations. In order to permit ready comparison between the spectra of the various fluctuations, we have in fact scaled all spectral intensities by dividing out appropriate polarization multipliers: Fig. 2a presents $N^2 I_1/I_3$ times the PSD of s_h, Fig. 2b presents N^2 times the PSD of s_3, and Fig. 2c presents I_1/I_2 times the PSD of v_3. All input spectra, thus scaled, are identical to those of Fig. 1.

The crude evaluation in H01 that led to Eqs. (3.1) and (3.2) for v_h is unable to distinguish between the various fluctuating quantities, aside from the same scalings. Accordingly, it is represented in Fig. 2 by the same dots as before. The dots therefore permit a ready means of comparing the Eulerian nonlinear responses of the various fluctuations relative to one another.

Perhaps the most notable feature of these new graphs is that found in Fig. 2b for s_3 at the smaller values of Ri_L. Specifically, the asymptotic tail for $Ri_L = 4.5$ lies below that for $Ri_L = 8.0$. Thus, while the coefficient of the m^{-3} term generally increases (as before) with decrease of Ri_L, in the case of s_3 it passes through a maximum at an Ri_L

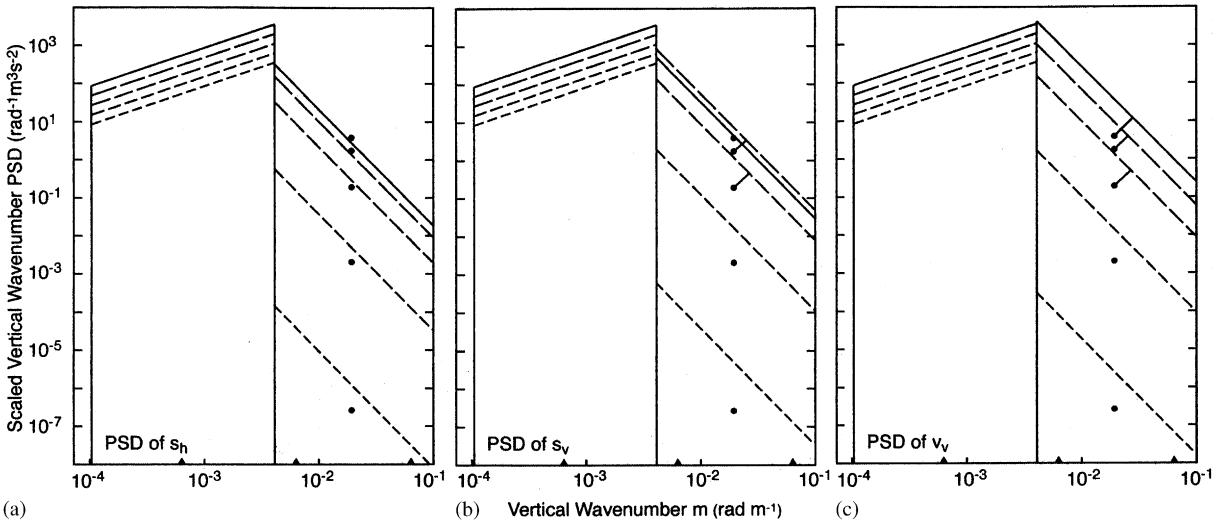

Fig. 2. As in Fig. 1, but now drawn for scaled PSDs of horizontal displacement s_h in panel (a), vertical displacement s_3 in panel (b) and vertical velocity v_3 in panel (c). The scaling is as described in Section 3, in order to match vertical scales with those of Fig. 1.

value somewhere between 8.0 and 4.5 and then decreases with further decrease of Ri_L. This behavior presents, on its own, an appearance of saturation within this range of Ri_L. It would be, however, saturation only of the s_3 spectrum, and so cannot account for saturation of the v_h spectrum.

(This manner of "saturation" had been anticipated in H01 from the form of Eq. (3.2) here, which reaches a maximum of 0.12 at Ri_L=3. Speculation was presented in H01 that such "saturation" might or might not be confirmed by numerical results. Evidently, confirmation is found for s_3, albeit at Ri_L somewhat larger than 3, but has not been confirmed for the other wave variables. Our numerical results were limited to $Ri_L > 3$, largely because the analysis in H01 is in the process of losing validity even as $Ri_L = 4.5$ is approached from above. The surprising behavior found here with s_3 might even be a result made false through neglect of the onsetting Lagrangian nonlinearity. We consider it to be irrelevant to the problem of saturation as it is observed in the middle atmosphere, and we discuss it no further. Ch02, p. 1766, accords his corresponding "saturation", found as if from Eq. (3.2) here, greater credence in this regard.)

The most significant comparison to be made is that of the PSD of s_3 in Fig. 2b with the PSD of v_h in Fig. 1b. This is because, as already noted, under present approximations the amplitude of horizontal velocity in each individual Lagrangian wave is given by N times the amplitude of vertical displacement in the same wave; hence the Lagrangian spectrum of v_h necessarily equals N^2 times the Lagrangian spectrum of s_3. This equality, which occurs also in linearized Eulerian waves, has been employed on occasion to infer that the same should hold true in observations of the "waves" of the Eulerian spectral tail. There is some danger that observational data relating to vertical displacements (or to proxies such as density or temperature fluctuations) might be transcribed into "equivalent" v_h data on this basis. Comparison of Fig. 1b with Fig. 2b—perhaps aided by comparison of the computed tail spectra relative to the associated dots—reveals that such a transcription may be quite unjustified. This is but one example, a very explicit one, of the general fact that linear polarization relations are not applicable in the nonlinear Eulerian tail.

Table 1
Approximate and accurate values of Z as functions of Ri_L

Ri_L	Z_a	Z_{v_h}	Z_{s_3}
4.5	0.10	0.43	0.11
8.0	4.1×10^{-2}	0.10	0.13
14.2	4.4×10^{-3}	9.0×10^{-3}	3.2×10^{-2}
25.3	4.1×10^{-5}	9.0×10^{-5}	6.3×10^{-4}
45.0	5.1×10^{-9}	1.7×10^{-8}	1.4×10^{-7}

A numerical comparison is presented in Table 1, where Z_a is the H01 approximation given here by Eq. (3.2), Z_{v_h} represents the value of Z produced from the present v_h computations, and Z_{s_3} represents the corresponding multiplier (of m^{-3}, not N^2m^{-3}) produced from the present s_3 computations. This table reveals, first, the extent to which the approximate Z_a is or is not a good representation of the corresponding factor appropriate to v_h and/or s_3. It also reveals the extent to which the scaling of s_3 data (or proxy data) into "equivalent" v_h data, via N^2, breaks down. It is evident that allegedly equivalent v_h spectra obtained from observed s_3 spectra can underestimate the truly equivalent v_h spectrum by as much as a factor of 4 (when $Ri_L = 4.5$), but can overestimate the truly equivalent v_h spectra by a factor nearly as

large or even larger (when Ri_L is much larger). Observers should take note of this anomaly.

(A further and highly significant departure from linear theory should be noted in the present context. As is described by Ch02 (Eq. (38) and following text), frequencies found at a chosen wavevector **k** well out in the Eulerian tail bear no relation to the frequency appropriate to **k** in a linear system. Instead, they form a Gaussian spread about zero frequency with the spread defined by the variance of Doppler shifting $\langle \mathbf{k} \cdot \mathbf{v} \rangle^2$. Temporally varying observations in the tail portion of the spectrum might then be interpreted as revealing waves of very low frequency, when they would in fact be revealing nonlinear behavior in "waves" of much higher frequency.)

The most popular value of Z_{v_h} found in the observational literature of the middle atmosphere is $1/6 = 0.17$, introduced first, on their own theoretical grounds and with compatible data, by Smith et al. (1987). Tsuda et al. (1989) present data that suggest a value of 0.3, once spectra for the two horizontal directions are combined. These two estimates are representative not only of the values found in atmospheric observations, but also of the range of values. This range is depicted in Fig. 1b as a short vertical bar. Comparison with Table 1 suggests that the observations relate to systems in which $4.5 < Ri_L < 8.0$, a conclusion compatible with that obtained above on the basis of form alone. Observers may be heartened to note that, somewhere in this range, the standard transcription of s_3 tail spectra into "equivalent" v_h tail spectra would seem to be quite accurate. (Alternatively, somewhere in this range the transcription could give a factor of 5/3, such as that reported by Tsuda et al. (1991), from temperature measurements.)

4. Height variations of spectra, and saturation

We now consider height variations of a wave spectrum as it propagates upward through the atmosphere. Together with all others who have considered the problem, we conceive of these variations as being governed by some balance between the standard growth of wave amplitude with height and the dissipative effects that this growth engenders. Together with most others, we take the dissipation to be controlled by instability, and we employ as a measurer of incipient instability a large-scale Eulerian Richardson number Ri_E defined as the unperturbed N^2 divided by Eulerian shear variance.

The H01 analysis of the Eulerian tail ignores dissipation except as an add-on that awaits incorporation, much as the theoretical analysis of Kelvin–Helmholtz instability usually begins with a smoothly sheared background flow that awaits further developments. With dissipation ignored, the m^{-3} form of the Eulerian v_h spectrum implies an m^{-1} form for the vertical shear spectrum. This, if integrated to infinite m, would yield infinite shear variance: Ri_E would be reduced to 0. The analytic results would be rendered meaningless by uncontrolled instability everywhere.

The present analysis assumes, therefore, that the Eulerian spectrum extends only to some terminal m, say m_T, such that Ri_E is of order 1. Its precise value would be determined in the middle atmosphere by a requirement that the associated instability be just sufficient to maintain an equilibrium via a leaching of wave energy as rapidly as it becomes available to the dissipating portion of the wave system.

We conceive of localized regions in physical space within which a locally defined Eulerian Richardson number (employing a wave-modified N^2 and the local shear) may drop to negative values, thereby inducing intense localized loss of wave energy. Such regions have been detected in the middle atmosphere and illustrated observationally by Sica and Thorsley (1996). We expect that the loss will be preferentially from the smallest-scale portions of the wave system, those that define and so correlate with the localized regions themselves. The resultant turbulence is also likely to remove wave energy from the smallest-scale structures preferentially, at least if its diffusivities (viscous and conductive) behave anything like molecular diffusivities. Accordingly, we expect the dominant effect of instability to be an erosion of the large-m edge of the Lagrangian spectrum as the waves propagate upward through the atmosphere. (Note that instability and turbulence extract energy from the physical wave system, whether that system be described in Eulerian or Lagrangian terms. Conditions for the occurrence of instability under gravity are best described in the Eulerian system, in which the direction of gravity is fixed, while the small-scale waves from which the energy is lost are best described in the Lagrangian system, since only in that system can they be identified individually as waves.)

To establish a base of discussion, we now take our model input spectrum to be of such low intensity, in an imagined region below the middle atmosphere, that $Ri_L = 45$. As is seen in Fig. 1c, the Eulerian spectrum will differ negligibly. Even the tail spectrum will contribute negligible shears, for any reasonable m_T, because of the six-or-so orders of magnitude drop of intensity. Ri_E will then differ negligibly from Ri_L, and so will approximate to 45. In such a wave field, a locally defined Ri_E would drop to 1 or less only in rare patches, here and there, now and then. (Small whitecaps on a slightly choppy sea provide a mental image.) Dissipation may occur in such regions, but we suppose it to be negligible for purposes of developing our picture.

The entire wave spectrum, being free of significant dissipation, amplifies with height above the imagined input level. Its Ri_L decreases because of its increasing shear variance, and its Eulerian spectral tail comes to take on significant magnitude. Ri_E decreases in tandem with Ri_L, but at a greater rate because of the additional shears associated with the Eulerian tail. The patches of incipient instability increase in size, dissipation comes to be significant, and erosion of the large-m end of the Lagrangian spectrum begins. Up to this point, the spectra depicted in Fig. 1c may be thought of as a sequence of spectra imagined at increasing heights, at least through the sequence $Ri_L = 45$, 25.3 and 14.2, and on to

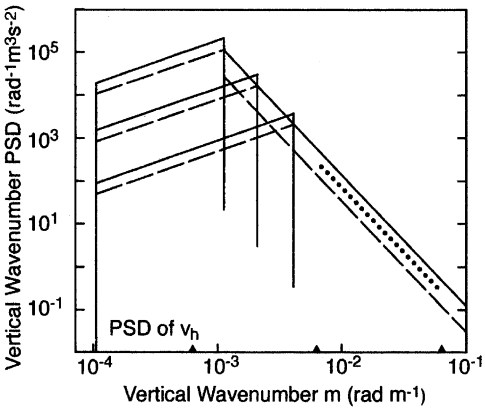

Fig. 3. Vertical wavenumber spectra of horizontal velocity v_h drawn for $Ri_L = 4.5$ (continuous sloping lines) and $=8.0$ (broken sloping lines), as in Fig. 1b, but now obtained for three different maximum input wavenumbers (with input intensities adjusted as required to provide constancy of the Ri_L values). The figure is designed to represent the growth of input intensity with height in combination with erosion of the small-scale portion of the input spectrum under dissipative losses. The constancy of output tail intensity that is found may be interpreted as saturation. The dotted line represents a typical middle-atmosphere tall intensity, with $Z = 0.2$.

$Ri_L = 8.0$ or thereabouts. Ri_E will have decreased even more, but we cannot assert how rapidly for want of knowledge of an effective m_T and its possible change with Ri_E.

We visualize an equilibrium state, as we enter the middle atmosphere, in which the large-scale Ri_E has decreased to some value of order 1 and the associated Ri_L has decreased to some larger value—perhaps 8.0 or 4.5, as illustrated. We assume that no further changes in these parameters occur: instability and resultant turbulence simply remove energy from the spectrum at a rate that maintains these values constant as height increases. A postulate similar to this is made, implicitly if not explicitly, in all other models of gravity-wave saturation in the middle atmosphere. What is crucially different here is that the nonlinearity of the Eulerian system has already automatically provided the form of the tail and a close estimate of its intensity, without additional postulates.

The assumption of a constant Ri_L and an associated erosion of only the large-m end of the Lagrangian spectrum leads to Fig. 3. This repeats the $Ri_L = 4.5$ and 8.0 cases illustrated in Fig. 1b, and it goes on to exhibit the corresponding spectra after growth of A with height and accompanying diminution of α_M, as implied by Eq. (2.7), Ri_L being held constant at its two illustrative values. The new spectra are drawn for heights where α_M has been reduced to (2π radians)/(3000 m) and (2π radians)/(6000 m) in turn. From Eq. (2.7), it can be seen that each such shift requires A to increase by a factor of 16, which corresponds to an increase in height of 2.8 atmospheric scale heights.

The essential point to note is that the new m^{-3} tail spectra have intensities almost identical to those already obtained: saturation with increase of height has been produced. This is the most important single conclusion of the present paper, once a general compatibility of tail intensity with observations has been established.

This manner of saturation was anticipated in H01 on the basis of the approximate Z_a given by Eq. (3.2), since that approximation depended only on Ri_L. Here it has been confirmed on the basis of the full numerical integrations: the accurate Z, though far more complex and dependent on all spectral sums, is also found to be virtually a function of Ri_L alone under changes of cutoff wavenumber α_M. (The same conclusion was found to apply to variables other than v_h, not shown, and it applies also at greater values of Ri_L than those shown here.) Moreover, as has been seen already, the magnitude of the accurate Z is fully compatible with the range of magnitudes observed, when Ri_L lies within the range 4.5 –8.0. This range would produce Ri_E values approximating to 1 if m_T were of order 100 times the m at the peak of the Eulerian spectrum, which is not an unreasonable value. (See sample calculation in Section 7.) Compatibility with observation is essentially complete.

The saturation obtained here can be attributed to the remarkable pattern of behavior of spectral sums and their subsequent combination described at the end of Section 2, a pattern that is inherent in the Lagrange $\rightarrow$ Euler transition. Had α_m been made to decrease along with α_M in our computations, saturation would have been achieved exactly. Fig. 3 contains a departure from this exact saturation, but one that is imperceptible to the eye—a consequence of the irrelevance of α_m when it is $\ll \alpha_M$.

Our spectrum was not chosen in advance to produce this result. Indeed, the remarkable behavior of spectral sums in combination was discovered almost accidentally in the course of the present study, after early results had become available. Moreover, the behavior itself will recur with spectra that are different from ours in form if only they satisfy the same scaling rule. The model spectrum adopted by Ch02, for example, would produce exact saturation if its horizontal and vertical characteristic wavenumbers were scaled identically, keeping Ri_L fixed. Departures from exact saturation (under constant Ri_L) with other chosen or observed spectra can arise only to the extent that those spectra fail to scale as described. One consequence of this behavior concerns our method of terminating and eroding the model Lagrangian input spectrum at large wavenumbers: a somewhat gradual rather than abrupt termination could have been assumed, with erosion over its whole range of large wavenumbers (as some would prefer and we would readily admit will occur in practice). With suitable scaling included, exact saturation would result; with slight departures from this scaling, only slight departures from exact saturation would arise.

5. Intense Eulerian and Lagrangian nonlinearity

The analysis of H01 depends upon the use of linearized Lagrangian relations to infer nonlinear Eulerian behavior.

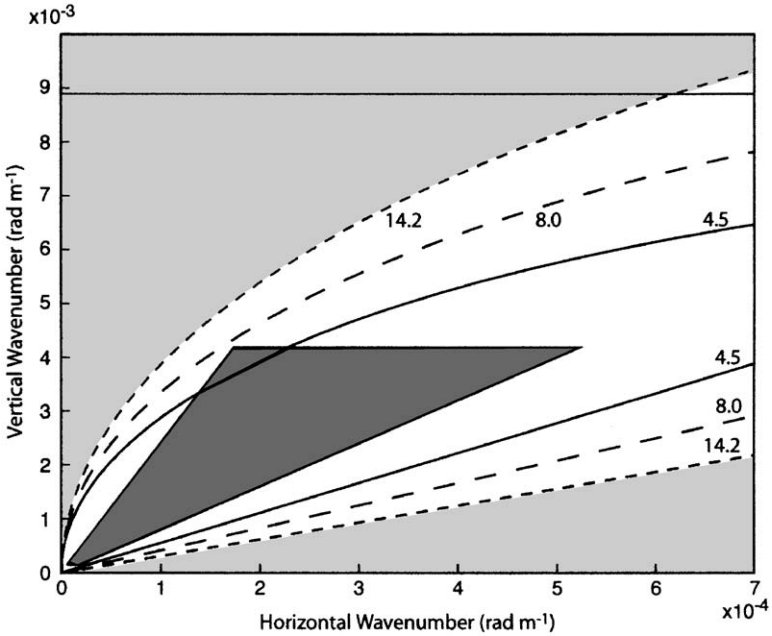

Fig. 4. Horizontal and vertical wavenumber space. Our model spectrum is confined to the central shaded area in the case $\alpha_M = (2\pi \text{ radians})/(1500 \text{ m})$. Intense Eulerian nonlinearity is to be found in the lightly shaded area at top left, and intense Lagrangian nonlinearity in the lightly shaded area at bottom right, when $Ri_L = 14.2$. The boundaries of these domains move to new locations as indicated, when Ri_L is reduced to 8.0 and then 4.5. See text for horizontal line.

There are those who would claim that, if the Eulerian equations demand nonlinear treatment, then the Lagrangian equations must also demand nonlinear treatment. This claim was refuted in H02, where distinctive domains of nonlinearity were found and illustrated. Certain quantities had to be $\ll$ other quantities for a linear treatment to be reasonably valid, and not all of the gross inequalities were equivalent in the two systems. We turn now to consider degrees of nonlinearity in the two, as they are found to occur in our spectra.

The discussion will be sharpened by introducing an abrupt change of nomenclature at points in parameter space where the gross inequalities come to be replaced by equalities: when the intensity of the wave system is less than that required to produce these equalities, the relevant waves will be termed "quasi-linear", and when greater they will be termed "intensely nonlinear". These terms are not intended to prejudge the role of nonlinearity in quasi-linear waves, but rather to make what is likely to be a useful distinction.

One condition on an Eulerian wave having wavevector $\mathbf{k}$, for it to be validly linearized, is well known to be that its frequency ω should be large in comparison to the Doppler shifting it undergoes by virtue of the $\mathbf{v} \cdot \mathbf{k}$ term imposed by the total velocity $\mathbf{v}$ of all waves. H02 introduced a statistical criterion appropriate to present circumstances, one that employed the velocity variance (in the form of its square root, the rms velocity): see its Eq. (2.9). Upon replacement of the gross inequality in it by an equality, that criterion leads to a transition from quast-linearity to intense nonlinearity when, in present notation,

$$\omega^2 = 0.5k_h^2 I_4 + k_3^2 I_5. \tag{5.1}$$

(The factor 0.5 arises because the chosen wave "sees" only the component of wind in its own direction of propagation, and this will account for only half the total horizontal-wind variance I_4.)

This transition is illustrated by curves in Fig. 4 for three values of Ri_L, in superposition on a depiction of our adopted spectral region when $\alpha_M = (2\pi \text{ radians})/(1500 \text{ m})$. The adopted spectrum, which would be equally an Eulerian and a Lagrangian spectrum at sufficiently low intensity, is confined to the central shaded region. The lightly shaded region at top left represents, for $Ri_L = 14.2$, the domain within which an Eulerian wave (specified by horizontal and vertical wavenumbers k_h and k_3) would be subject to intense nonlinear interaction, to the point that it could hardly be reasonably thought of as an identifiable wave at all. The remainder of the k_h, k_3 space, below the shading and to its right, would represent quasi-linear waves insofar as this source of nonlinearity is concerned. The domain of intense nonlinearity encroaches on the quasi-linear domain as the intensity of the spectrum is increased, as is depicted by adjacent boundary curves drawn for $Ri_L = 8.0$ and then 4.5. When $Ri_L = 4.5$, it even encroaches on the domain of the input spectrum. The overlapped portion of that spectrum

cannot possibly be considered quasi-linear in the Eulerian system.

The onset of the overlap occurs when Ri_L is reduced to 7. The transition seems consistent with our previous figures: when Ri_L substantially exceeded 7, the inferred Eulerian spectra were closely similar to the Lagrangian spectra, with the tail producing only a relatively minor addition, but the departures of the Eulerian from the Lagrangian spectra became major for $Ri_L < 7$. The transition criterion seems to serve its purpose reasonably well.

(It is worth noting that the boundary curves asymptote to horizontal lines at large horizontal wavenumbers, the horizontal line in Fig. 4 being that for $Ri_L = 4.5$. This line represents the place where the boundary curve would have been, if Doppler spreading were attributed to horizontal winds alone. The wave-induced vertical velocity is seen to be as significant as the wave-induced horizontal velocity is, in producing intense nonlinearity at the relevant large vertical wavenumbers. This fact should be kept in mind by observers, when comparing the onset of the tail in observed spectra with the point at which the onset of intense nonlinearity is to be expected.)

A Lagrangian description of the wave system avoids the foregoing nonlinearity entirely. There is, however, a replacement nonlinearity that is dependent on the vertical shear of horizontal displacement. For it, the transition from quasi-linearity to intense nonlinearity occurs when $(\alpha_3/\alpha_h)^2$ equals $0.5I_3$, as may be inferred from Eq. (3.22) of H02. The corresponding domain of intense nonlinearity is shown, lightly shaded, in the bottom right corner of Fig. 4 for $Ri_L = 14.2$. The remainder of what is now α_h, α_3 space—above and to the left—represents waves that are quasi-linear in the Lagrangian system, insofar as this nonlinearity is concerned. As before, the domain of intense nonlinearity encroaches on the domain of quasi-linearity as Ri_L is reduced, as is illustrated by new boundaries for $Ri_L = 8.0$ and 4.5. In this case, however, there is no encroachment on the domain of the adopted spectrum: the latter continues to represent quasi-linear Lagrangian waves throughout our analysis. (Encroachment would require that Ri_L be reduced to 2.2.) This is the crucial distinction that underlies the use of linear Lagrangian spectra to infer nonlinear Eulerian spectra, in the formulation of H01 and so in the present work.

There is another source of nonlinearity that affects Eulerian and Lagrangian waves almost equally, associated with low frequencies: see Eqs. (2.19) and (3.21) of H02. It would be represented in Fig. 4 by a triangular region adjacent to the vertical axis and encroaching on the quasi-linear domain from there, mirroring the encroachment that comes from the bottom right. It would not reach the domain of the adopted spectrum until Ri_L was reduced to about 2, and so never does in our work.

Finally, each system necessarily transits to an intensely nonlinear state if its own Richardson number decreases to 1. This never occurs in our Lagrangian work, so again there is no intense Lagrangian nonlinearity. The same would be true of the Eulerian system if the evaluation were made for the input spectrum itself, and so that spectrum would not be found to suffer intense nonlinearity on this basis.

Related to the question of nonlinearity is the assumption made in H01 that the Jacobian of the transformation between Eulerian and Lagrangian coordinates could be taken as 1. This was done purely as a convenience: had it not been done, the analysis would have required an order of magnitude increase of effort, well beyond anything manageable, with no change of end result save in minor details. With our chosen linearized Lagrangian waves, there are in fact nonlinear departures from $J = 1$. They consist of terms approximating to I_1 and $(I_2 I_3)^{1/2}$, and so rise to values approximating 0.2 as Ri_L falls to 4.5 in our work. This adds to the reservations we must associate with detailed results for the smallest Ri_L values, but it in no way undermines the trend toward those results or their general validity.

In summary, the Eulerian system exhibits intense nonlinearity in circumstances in which the Lagrangian system remains quasi-linear, namely $2 < Ri_L < 7$. The difference arises because of the Doppler shifting that the Eulerian system introduces, a shifting that is absent from the Lagrangian system and whose replacement nonlinearity acts quite differently. Of course, we have designed our input spectrum in part to achieve precisely this end, for pedagogic purposes, and we could have generated an input spectrum that would have reversed the roles of Eulerian and Lagrangian non/linearity. But the Eulerian Doppler-shift nonlinearity is an admitted feature of waves in the middle atmosphere, so its consequences are the ones we have sought to probe and display here.

Even in our chosen circumstances, with the Lagrangian system quasi-linear, one may object by saying that the Lagrangian system is, equally, quasi-nonlinear. We are undoubtedly extrapolating a linearized Lagrangian analysis into a substantially nonlinear regime, but even this is unlikely to produce serious error. The fields of all fluctuating parameters (such as displacement and velocity components, here) can certainly be Fourier analyzed in Lagrangian space according to wavevector $\boldsymbol{\alpha}$ at any given time. All that is really required for validity of the 3D portion of the Lagrange $\rightarrow$ Euler transformations of H01—and so, too, for the validity of the present study—is that the ratios of the various parameters to one another, expressed as functions of wavevector components, should not differ greatly from those found in linear theory. Just such a behavior is commonly attributed to weakly interacting waves, the interactions being absorbed into a set of slowly varying complex amplitude factors, one for each $\boldsymbol{\alpha}$ in turn. This behavior implies, then, that the sequences of change exhibited here (with decreasing Ri_L) are robust under the onset of Lagrangian nonlinearity and into its early stages, such as we have here at the smallest Ri_L values. Since the Eulerian system is intensely nonlinear in the related circumstances of interest, we believe that extrapolation well into the realm of Lagrangian quasi-nonlinearity is fully justified as a means of determining Eulerian results.

In fact, it may well be the only means. That it has yielded results consistent with observations can hardly be denied.

6. Comparison with DST

The DST of Hines (1991, 1993) was an attempt to evaluate the role that might be played by the Eulerian $\mathbf{v} \cdot \nabla$ nonlinearity in the production of vertical-wavenumber tail spectra. It mathematically developed a somewhat heuristic approach, and for this reason it and its offspring DSP have been subject to legitimate doubts. We have available here the results of a far more rigorous analysis, against which those of the DST can be tested.

The DST assumed an input Eulerian spectrum of sufficiently low intensity that nonlinearities could be ignored, and then sought to determine the nonlinear effects as intensity increased. The input spectrum played a role, relative to the pending nonlinear output, identical to that of the present Lagrangian input spectrum. We therefore insert our prototypical v_h input spectra—those leading to Fig. 1b—into the DST formula for the output Eulerian spectrum. This yields, from Eq. (2.18) of Hines (1991),

$$P_E(m) = \pi^{-1/2} N \sigma^{-1} m^{-1} \int P_L(m_i) m_i^{-1} \times \exp[-(M_i^{-1} - M^{-1})^2]\, dm_i, \qquad (6.1)$$

where P_L is the input vertical-wavenumber (m_i) Lagrangian PSD, P_E is the output vertical-wavenumber (m) Eulerian PSD, $M_i = m_i\sigma/N$, $M = m\sigma/N$, and σ^2 is the variance of v_h, given by I_4 here.

A comparison is made between the DST output spectra and the newly derived spectra in Fig. 5, for the two most relevant of our chosen Ri_L values, namely 4.5 and 8.0. (The spectra for $Ri_L = 8.0$ have been reduced by two orders of magnitude for clarity.) As will be seen, the DST correctly represents the spectral peak. It further gives a fair representation of the onset of the tail, particularly in the case $Ri_L = 8.0$. Indeed, omission of the trend to the admittedly erroneous m^{-1} spectrum, and substitution of an m^{-3} tail as initially advocated, would produce a net spectrum in excellent agreement with the far more rigorous result. The utility of the DST analysis as a means of estimating the intensity of the asymptotic tail, via a relatively trivial calculation (in contrast to the present, tortuous route), would appear to have been established at the relevant values of Ri_L. The comparisons exhibit progressive deterioration of the agreement as Ri_L is increased beyond 8 (not shown), though the trend toward greatly reduced tail intensities is as clear in the original DST results as in our new spectra.

One should not read too much into the favorable comparison. For one thing, identical DST curves would have been found, after appropriate scaling, for s_h, s_3 and v_3, and we have already seen that these variables produce m^{-3} tails that are not scaled identically. Nevertheless, for the variable of greatest observational interest in the middle atmosphere—the horizontal velocity—the agreement appears to be quite satisfactory on an empirical basis.

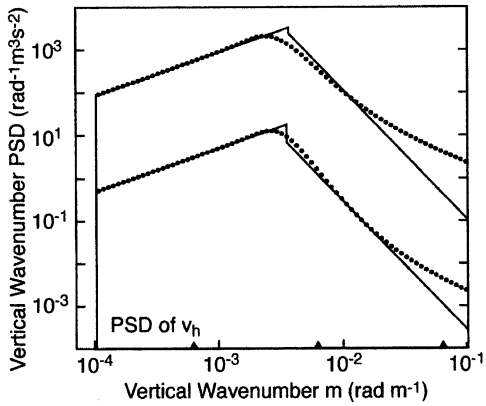

Fig. 5. Input and output spectra as in Fig. 1b for $Ri_L = 4.5$ and 8.0 (with the latter reduced by 10^2 for clarity; continuous and broken sloping lines, respectively), together with corresponding output spectra (dotted curves) that are produced by the Doppler-spread relation (6.1). The DST advocated use of (6.1) up to wavenumbers such that an m^{-1} form was found to be setting in, and thereafter an m^{-3} form.

A more thorough discussion of the DST and of its offspring DSP, as viewed in the light of these results and their foundations, is given in a companion paper (Hines, 2004), together with a possible upgrading of the DSP.

7. Discussion

There is widespread agreement that a broad spectrum of gravity waves ascends through the middle atmosphere. Also, at large vertical wavenumbers the Eulerian m spectrum has roughly the form ZN^2m^{-3} with Z roughly constant in height. Observational constraints preclude assessment of the m spectrum at m values markedly less than those of the m^{-3} tail, but its existence is revealed by frequency spectra. These exhibit increase of total spectral intensity (as measured by the irregular-wind variance σ^2) by a factor of 100 or more from the tropopause to the mesopause (e.g., Balsley and Carter, 1982; Balsley and Garello, 1985). Since the tail is not growing with height, the sub-tail portion of the spectrum must be. This portion is also widely held to be the energy source of the tail structures at higher elevations, the tail portion of the spectrum having moved gradually into and having eroded the large-m part of the sub-tail portion as height increased.

From these considerations, it is widely held that dissipation of the wave system occurs predominantly in those portions of it that provide the tail. This dissipation is accompanied by deposition of momentum, which in turn is important in driving the middle-atmospheric circulation. An understanding of the mechanisms of formation and dissipation of the tail is therefore essential if a valid physically

based parameterization of the deposition is to be achieved for purposes of numerical modeling.

One can compare the m values found in the Eulerian tail at any given height with the rms speed σ of the irregular winds observed at that height. These m values are invariably centered near N/σ, a fact which automatically implies that the "waves" of the tail are undergoing significant nonlinear interactions if the spectrum is indeed broad. We have shown that, simply by recognizing this fact and assessing the implications, we produce an m^{-3} tail at observed intensities if Ri_{L} is in the range 4.5–8.0. We have shown also that this tail retains its intensity roughly unchanged as height increases—i.e., as the intensity of the sub-tail portion of the spectrum increases—provided only that Ri_{L} remains roughly unchanged as a consequence of erosion of the large-m Lagrangian waves; and we have argued that Ri_{L} is likely to remain unchanged as a consequence of Ri_{E} remaining unchanged at some value of order 1 in conformity with a widely accepted condition for incipient instability of the spectrum.

These conclusions support the concept of the Eulerian tail being a consequence of the nodissipative $\mathbf{v}\cdot\nabla$ Eulerian nonlinearity, as developed initially in the DST and as further discussed in the present companion paper. They also serve to introduce severe constraints on most alternative models of tail formation, saturation and momentum deposition—those that treat the "waves" of the tail as if they were validly linearized and subject to the linear dispersion and polarization relations. The argument falls into two parts, depending on whether the alternative model accepts that the spectrum is broad and present at observed intensities continuously while the "waves" of the tail are analyzed, or instead that the "waves" of the tail are present only individually or in individual wave packets at any point of interest at the time of interest.

In the former case, one adopts some model broad m spectrum for the horizontal velocity, of which a typical example would be $ZN^2m_*^{-4}m(1+m^4/m_*^4)^{-1}$. Here, m_* is a characterizing wavenumber that marks the onset of the tail: if the spectrum is plotted in log–log coordinates, the asymptotes defined at small and large m would, when projected, intersect at $m=m_*$, while the actual spectral maximum occurs at $m=3^{-1/4}m_*=0.76m_*$. (The Eulerian spectrum shown for $Ri_{\mathrm{L}}=4.5$ in Fig. 1c serves to illustrate this spectral form, having been drawn according to its prescription with asymptotes matched to those in Fig. 1b.) The corresponding shear spectrum is found via multiplication by m^2. Integration of both spectral forms is readily accomplished. It yields a horizontal-velocity variance $\sigma^2=0.25\pi ZN^2/m_*^2$ when integrated to $m\gg m_*$, and an Eulerian Richardson function $Rif_{\mathrm{E}}(m)=4(Z\ln[1+m^4/m_*^4])^{-1}$ when integrated to some general m. (Rif_{E} is defined like Ri_{E}, but employs only the shear variance derived from wavenumbers less than the m of current interest.) The tail is taken to terminate at some m_{T} such that Ri_{E} is of order 1. For $Ri_{\mathrm{E}}=1$ and $m_{\mathrm{T}}\gg m_*$, this requires $m_{\mathrm{T}}=m_*\exp[Z^{-1}]$. The components of this spectrum—including the components that make up its tail—are then treated as if they behave like linearized Eulerian waves for the calculation of momentum flux, etc., at least as an approximation.

This model is subject to a *reductio ad absurdum* argument of the following form.

We adopt, for purposes of illustration, the observationally inferred $Z=0.2$. This produces $\sigma=0.4N/m_*$ and, for $Ri_{\mathrm{E}}=1$, $m_{\mathrm{T}}=148m_*$. But linearized gravity waves have horizontal phase speeds that approximate N/m, so all spectral components at $m>m_\sigma\equiv N/\sigma=2.5m_*$ have horizontal phase speeds less than the rms wind speed generated by the wave system. Even proponents of the broad-spectrum model will concede that the corresponding "waves" must be affected by nonlinear interaction with other waves. Pursuit of the model must therefore depend on a hope that the nonlinear effects will not seriously distort the use of linearized approximations. This hope cannot be supported.

Let us grant, for purposes of argument, that the spectrum can indeed be approximated as linearized Eulerian waves for $0<m<m_\sigma$, but we momentarily withhold judgment of the spectral content at larger m. Given such an assumption, the waves in this range can be described equally well as linearized Lagrangian waves having the same spectrum. The Lagrangian Richardson number Ri_{L} for this range can then be obtained from Rif_{E}, by taking $m/m_*=m_\sigma/m_*$. It is 5.4. But, as seen above, such an Ri_{L} already produces an m^{-3} Eulerian tail with Z somewhat greater than 0.2, and so already accounts for all the spectral content lying at $m>m_\sigma$. There is simply no room for the addition of further independently propagating, quasi-linear Eulerian waves. The spectrum beyond m_σ cannot consist of such waves at all, but only of deformations imposed on the smaller-m waves by one another (as is described by Eckermann, 1999). The smaller-m waves simply propagate as they choose to do, and the Eulerian "waves" beyond m_σ must adapt. The latter have no claim whatever to an independent existence. Instead, they are just spectral components continuously created and modified at will by smaller-m waves. See Ch02 for further discussion of the process.

Indeed, the requirement for equality of areas under the Eulerian and Lagrangian spectra implies that our earlier concession was too generous. If the model correctly presents the Eulerian spectrum (as assumed), then the Lagrangian spectrum must have exceeded the Eulerian even at $m<m_\sigma$ (to equalize areas) and must have been terminated or grossly reduced well before m_σ is reached, leaving even more of the Eulerian tail as nothing but the creature of the smaller-m waves. Consequences such as this follow as a mathematical imperative of the Lagrange $\rightarrow$ Euler transformation.

This imperative implies that models of middle-atmosphere wave systems that entail the simultaneous presence of many waves from many sources cannot properly represent momentum flux in the tail via linearized Eulerian analysis. Of course, parameterizations are often forgiving, so any that are based on such analysis may nevertheless manage to be successful in much of their output. But they

should always be held in suspicion, in the knowledge that the physics underlying them is not valid: they may, in some new circumstances, lead to quite erroneous results for that reason.

It appears that the only escape from the mathematical imperative lies in a hope that the observed waves of the tail derive from wave packets that propagate without overlapping one another to any substantial degree, as in the second model. This would assume them to be propagating much like bullets shot randomly into the air from various sources, all bullets being of such small cross section that they do not run into one another but nevertheless continuously fill the observer's field of detection. It further assumes an absence of sub-tail waves, despite the need for them in order to produce tail waves at greater heights. Any such picture should be justified by its proponents on the basis of estimated wave-packet sizes and durations that are compatible both with mathematical constraints and with realistic intensities for the wave system. Estimates of these intensities must include the full-wave system, not just the portion found in the local tail. Such a model seems likely to be found quite untenable when put to this test. It certainly departs from the view adopted by most observers in interpretations of their data.

In contrast, whatever their faults, the DST and DSP at least work compatibly with the mathematical imperative—even more acceptably, perhaps, when revisited and revised as in the present companion paper.

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Appendix A. The discarded m^{-1} term (by C.O.H.)

This paper presents Eulerian spectral tails having the form m^{-3}. The literature has from time to time implied the existence of tails having the asymptotic form m^{-1} as $m \to \infty$. An anonymous reviewer has requested elaboration of certain aspects of the appearance and elimination of this asymptotic form.

The form first appeared in the present context in the heuristically based DST of Hines (1991). It was immediately discredited in that same paper as being a consequence of inadequate approximations, some of which were then discussed. In its place, attention was directed to the quasi m^{-3} form that was found in a transitional portion of the spectrum, and this form was adopted for further analysis. The m^{-1} form did appear in later discussion of the DST, notably in criticisms of it, but rarely to useful purpose. This particular m^{-1} form may be dismissed out of hand in the future, without, however, dismissing the context in which it was found.

Hines (1991) accepted as a far more rigorous approach to the same problem the analysis by AJ89, which led to the expected m^{-3} form. Oddly enough, both Ch02 and H01 subsequently found, in separate reexaminations and redevelopments of AJ89, that that analysis had overlooked a term that should have produced the form m^{-1} under its own protocols. This led to a dilemma, in that the analysis appeared completely sound and yet the m^{-1} term had to be wrong, if only to maintain finite Eulerian velocity variance given finite Lagrangian velocity variance. The relevant term had been overlooked by AJ89 through the use of periodic boundary conditions in an integration by parts, in company with an improper invocation of the cancellation of end-point terms that usually arises in such circumstances. (See Appendix B of H01 for the nature of the mathematical impropriety. More physically: the AJ89 analysis depended upon two-point correlations that fell rapidly to zero as the separation distance increased, but it then ignored the fact that the use of periodic boundary conditions implied full correlation between points at opposite sides of the bounded region. Unphysical though this correlation may be thought, its omission as a part of the mathematics is not valid.)

Ch02 avoided the resultant dilemma simply by restricting discussion to intermediate m values such that the m^{-3} term would dominate over the m^{-1} term, and by noting that untreated problems of Lagrangian nonlinearity would in any event arise at greater m values. Only the m^{-3} term was given detailed study as being relevant to the observed range of m.

H01, on the other hand, pursued the matter further to find an appropriate means of negating the new m^{-1} term. Apparent survival of the term into later stages was traced to the breakdown of a theoretical-physics approximation standardly used in the evaluation of correlations in a field of randomly superimposed fluctuations, when applied to the clear-cut case of the correlation of a constant with itself when the fluctuations are pure sinusoids. An exact evaluation would have integrated the m^{-1} term out of existence, as was shown, whereas the approximation did not. (See Appendix C of H01.) This conclusion established that the unwanted m^{-1} term was in the end quite spurious, beyond a doubt, in spectra of field variables that are in-phase with Lagrangian displacement in individual waves (such as displacement itself, studied here), and equally for phase-quadrature variables (such as velocity, also studied here) if the spectra are vertically symmetric (when in-phase behavior reappears). A similar conclusion can be inferred indirectly for phase-quadrature variables even in the presence of vertical asymmetry by application of the requirement for finite Eulerian variances (see Appendix B here). The source of the error in that case may be expected to lie in a similar breakdown of the standard theoretical-physics approximation, but formal proof of this has yet to be found.

There is no similar reason to expect the breakdown of the standard approximation for higher-order terms. However, as

stated in H01, "With the AJ89 protocol (i.e., the standard theoretical-physics approximation) now known to produce error when applied to (the correlation of a constant with itself), some suspicion of error in the coefficients of the higher-order terms may arise. We nevertheless proceed with the analysis assuming the validity of those coefficients in the retained lead term (i.e., the m^{-3} term), and recognizing the risk. The ultimate test of the utility of the resultant analytic relations lies in the insights to which they lead and in the degree to which observations may be interpreted with their aid". The computations presented in this paper and their compatibility with observation surely do encourage belief in the validity of the formulae on which they are based. Moreover, the caveat quoted here is one that is implicit, though left unsaid, in most approximations made in physical analysis, and certainly in the particular approximation at issue here: if the formulae derived from this approximation are incorrect here, then there is a vast body of published work in a variety of fields that is equally incorrect. No hint of such widespread error has yet surfaced.

Appendix B. Comparison of Eulerian and Lagrangian spectral integrals (by J.B.K.)

Suppose we have a scalar field (such as any component of $\mathbf{s}$ or $\mathbf{v}$, here) given functionally by $L(\mathbf{r})$ in Lagrangian coordinates $\mathbf{r}$ and by $E(\mathbf{x})$ in Eulerian coordinates $\mathbf{x}$. Let $\mathbf{x} = \mathbf{f}(\mathbf{r})$ be the transformation relating these two reference frames, and assume for the time being that the Jacobian $J(\mathbf{r})$ corresponding to this transformation is identically 1 as a result of incompressibility.

Now imagine we measure E in some finite region ψ of the Eulerian coordinate system. Implicitly, we then know L over the corresponding region ϕ of the Lagrangian system, which is mapped to ψ by $\mathbf{f}$. The assumption $J = 1$ gives the obvious relation $\mathrm{vol}(\psi) = \mathrm{vol}(\phi)$. For the sake of mathematical clarity, let $E_\psi(\mathbf{x}) = E(\mathbf{x})$ when $\mathbf{x}$ is in the region ψ, and zero otherwise. Define L_ϕ similarly. The Eulerian power spectrum is then given by

$$\mathrm{PSD}[E_\psi] = (2\pi)^{-3}\mathrm{vol}^{-1}(\psi)|\widehat{E}_\psi(\mathbf{k})|^2, \tag{B.1}$$

where $\widehat{E}_\psi(\mathbf{k})$ denotes the Fourier transform of E_ψ. Similarly, the Lagrangian power spectrum is

$$\mathrm{PSD}[L_\phi] = (2\pi)^{-3}\mathrm{vol}^{-1}(\phi)|\widehat{L}_\phi(\boldsymbol{\alpha})|^2. \tag{B.2}$$

Plancherel's Theorem (alternatively known as Rayleigh's or Parseval's theorem) gives

$$\begin{aligned}\int |\widehat{E}_\psi(\mathbf{k})|^2\,\mathrm{d}^3\mathbf{k} &= (2\pi)^3\int |E_\psi(\mathbf{x})|^2\,\mathrm{d}^3\mathbf{x}\\ &= (2\pi)^3\int |L_\phi(\mathbf{r})|^2 J(\mathbf{r})\,\mathrm{d}^3\mathbf{r},\\ &= \int |\widehat{L}_\phi(\boldsymbol{\alpha})|^2\,\mathrm{d}^3\boldsymbol{\alpha}\end{aligned} \tag{B.3}$$

since we are supposing $J(\mathbf{r}) = 1$. Consequently,

$$\int \mathrm{PSD}[E_\psi]\,\mathrm{d}^3\mathbf{k} = \int \mathrm{PSD}[L_\phi]\,\mathrm{d}^3\boldsymbol{\alpha}. \tag{B.4}$$

That is, the integrals of the Eulerian and Lagrangian power spectra are identical when $J(\mathbf{r}) = 1$ identically.

Using similar arguments, we can see that if J varies only slightly from place to place, Eq. (B.4) should still be a good approximation. In general, if we have $0 < A < J(\mathbf{r}) < B$ everywhere, then $A\,\mathrm{vol}(\phi) < \mathrm{vol}(\psi) < B\,\mathrm{vol}(\phi)$ and

$$\begin{aligned}&(A/B)\int \mathrm{PSD}[L_\phi]\,\mathrm{d}^3\boldsymbol{\alpha}\\ &\quad < \int \mathrm{PSD}[E_\psi]\,\mathrm{d}^3\mathbf{k} < (B/A)\int \mathrm{PSD}[L_\phi]\,\mathrm{d}^3\boldsymbol{\alpha},\end{aligned} \tag{B.5}$$

although one would hardly expect the middle term to approach either of these bounds in practice unless A/B is itself close to 1. The assumption of incompressibility ($J = 1$) is therefore not mathematically precarious, at least in this context.

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